CME 307 / MS&E 311 / OIT 676: Optimization

Gradient descent

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Management Science and Engineering
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October 27, 2026

Outline

Unconstrained minimization

Quadratic approximations

Analysis via Polyak-Lojasiewicz condition

Unconstrained minimization

$$\text{minimize} \quad f(x)$$

- $f : \mathbb{R}^n \to \mathbb{R}$ differentiable
- assume optimal value $f^\star = \inf_x f(x)$ is attained (and finite)
- assume a starting point $x^{(0)}$ is known

unconstrained minimization methods

- produce sequence of points $x^{(k)}$, $k = 0, 1, \ldots$ with

$$f(x^{(k)}) \to f^\star$$

  (we hope)

Gradient descent

$$\text{minimize} \quad f(x)$$

idea: go downhill

Algorithm Gradient descent

Given: differentiable $f : \mathbb{R}^n \to \mathbb{R}$, $x^{(0)}$, stepsizes $t_k > 0$

For: $k = 0, 1, \ldots$

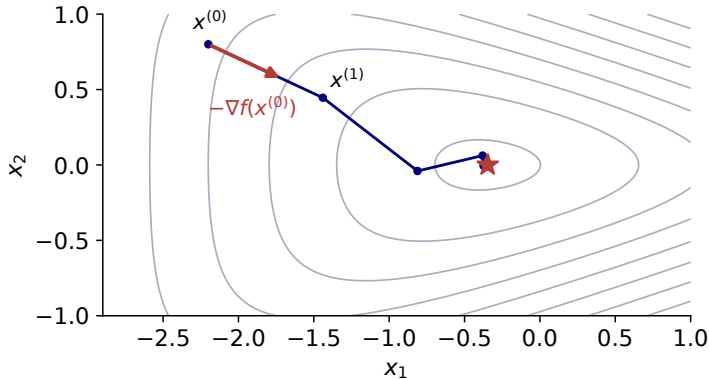
- stop if $\|\nabla f(x^{(k)})\| \leq \varepsilon$ (or iteration budget reached)
- update x:

$$x^{(k+1)} = x^{(k)} - t_k \nabla f(x^{(k)})$$

---

small gradient certifies approximate stationarity; when does it certify near-optimality?

Gradient descent: picture



each step moves along $-\nabla f$, which is orthogonal to the level set through the current point (here: backtracking line search on a smooth convex f)

Gradient descent: choosing a step size

- **constant step size.** $t_k = t$; how small is small enough?
- **decreasing step size.** $t_k \to 0$ but $\sum_k t_k = \infty$; for example, $t_k = t_0/(k+1)$
- **line search.** choose t_k so the new objective

$$f(x^{(k)} - t_k \nabla f(x^{(k)}))$$

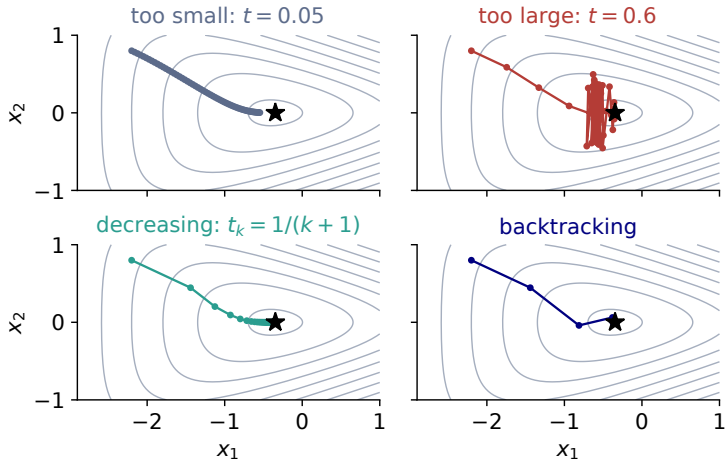
  decreases enough

tradeoff:

- a constant step can be too small (slow) or too large (diverge)
- decreasing steps guarantee $\nabla f(x^{(k)}) \to 0$[1] but converges slowly
- line search uses extra evaluations of f

[1] if ∇f is Lipschitz and f is bounded below

Step size rules: picture



60 iterations of each rule, with the same f and $x^{(0)}$ as the GD picture

Line search

write $g = \nabla f(x)$ and $x^+ = x - tg$

- **exact line search:** minimize $f(x - tg)$ over $t \geq 0$, if a minimizer exists
- **Armijo rule:** accept $t > 0$ if

$$f(x - tg) \leq f(x) - ct\|g\|^2, \qquad c \in (0, 1)$$

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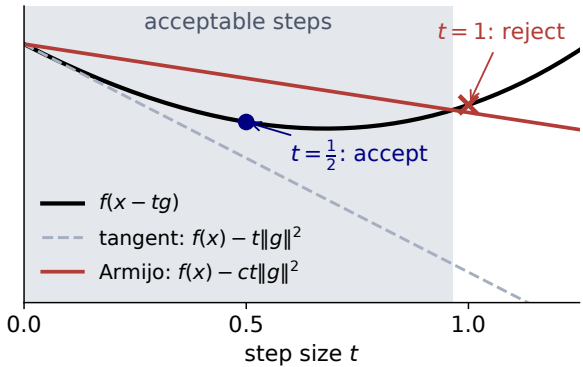
$$f(x - tg) \leq f(x) - ct\|g\|^2, \qquad c \in (0, 1)$$

Q: Can we always satisfy the Armijo rule for some $t > 0$?
A: Yes, for differentiable f: if $g \neq 0$, then as $t \downarrow 0$,

$$f(x - tg) = f(x) - t\|g\|^2 + o(t).$$

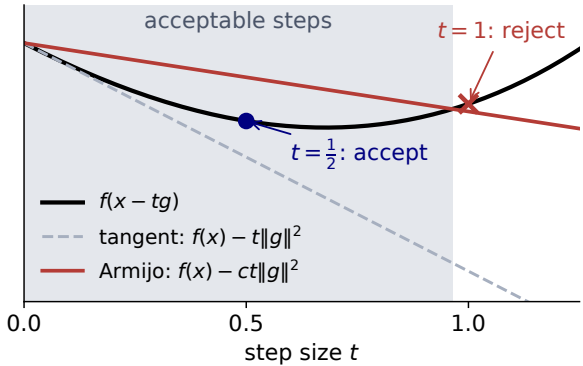
If $g = 0$, the rule holds with equality (and we can stop).

Backtracking: picture



shaded: steps where $f(x - tg)$ lies below the Armijo line (here $c = 0.3$, for visibility)

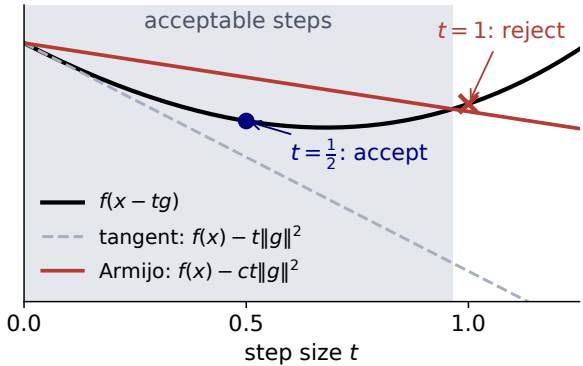
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Q: Why not use the tangent line $f(x) - t\|g\|^2$ as the threshold?

Backtracking: picture



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Q: Why not use the tangent line $f(x) - t\|g\|^2$ as the threshold?

A: For convex f, the curve lies above its tangent: only $t = 0$ would pass.

Demo: gradient descent

Open the gradient descent notebook

Q: Before running: what happens if we double the step size?

- predict: faster progress, oscillation, or divergence?
- compare a fixed step with backtracking
- record the objective value and gradient norm

Backtracking line search

choose $c \in (0, 1)$, for example $c = 0.01$

1. compute $g = \nabla f(x)$ once; set $t = 1$
2. while $f(x - tg) > f(x) - ct\|g\|^2$, halve t
3. accept $x^+ = x - tg$

Backtracking line search

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why does it terminate? sufficiently small steps satisfy Armijo

How well does GD work?

Open the contour notebook

- $f(x) = \|x\|^2$: try $t = 1/2$, $t = 0.95$, and $t > 1$
- $f(x) = x^T A x$, $A \succeq 0$: what changes when the contours stretch?

How well does GD work?

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What if the assumptions fail?

- $f(x) = \|x\|_1$: the gradient is undefined at some points
- $f(x) = 1/x$, $x > 0$: the infimum is not attained; iterates can escape to infinity

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Quadratic approximation

suppose $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable

$$f(y) \approx f(x) + \nabla f(x)^T (y - x) + \tfrac{1}{2}(y - x)^T \nabla^2 f(x)(y - x)$$

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$$f(y) \approx f(x) + \nabla f(x)^T (y - x) + \tfrac{1}{2}(y - x)^T \nabla^2 f(x)(y - x)$$

replace the Hessian by a **positive definite** matrix $H \succ 0$:

$$q_x(y) = f(x) + \nabla f(x)^T (y - x) + \tfrac{1}{2}(y - x)^T H(y - x)$$

$$\nabla q_x(y^\star) = 0 \quad \Longrightarrow \quad y^\star = x - H^{-1} \nabla f(x)$$

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- choose $H = \frac{1}{t} I$, $t > 0$, to obtain **gradient descent**:

$$x^+ = x - t \nabla f(x)$$

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- choose $H = \nabla^2 f(x)$ (if $\nabla^2 f(x) \succeq 0$) to obtain **Newton's method**
- choose another psd approximation of the Hessian to obtain a **quasi-Newton method**

Smoothness: bounded curvature

throughout this section, suppose $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable

Definition (Smooth)

f is L-**smooth**, $L > 0$, if for every x,

$$-LI \preceq \nabla^2 f(x) \preceq LI.$$

all Hessian eigenvalues lie in $[-L, L]$: curvature can be negative

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Q: When is $f(x) = \frac{1}{2}x^T Ax$ L-smooth? for what L?

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Q: When is $f(x) = \frac{1}{2}x^T Ax$ L-smooth? for what L?
A: Any $L \geq \max(|\lambda_{\min}(A)|, \lambda_{\max}(A))$ with $L > 0$.

The descent lemma

write $d = y - x$. Taylor's theorem gives, for some $\theta \in (0, 1)$,

$$f(y) = f(x) + \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(x + \theta d) d \, d$$

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since $\nabla^2 f(z) \preceq LI$,

Quadratic upper bound

$$f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} \|y - x\|^2.$$

Which step sizes guarantee descent?

write $g = \nabla f(x)$ and substitute $y = x - tg$ into the upper bound:

$$\begin{aligned} f(x - tg) &\leq f(x) - t\|g\|^2 + \frac{Lt^2}{2}\|g\|^2 \\ &= f(x) - t\left(1 - \frac{Lt}{2}\right)\|g\|^2. \end{aligned}$$

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Safe step sizes

If $g \neq 0$, then $0 < t < 2/L$ guarantees $f(x - tg) < f(x)$.

the downhill gain grows like t; the curvature penalty grows like t^2

Optimizing the upper bound

which safe step gives the largest **guaranteed** decrease?

$$t - \frac{Lt^2}{2} \quad \text{is maximized at} \quad t = \frac{1}{L}.$$

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this is also the step that minimizes the quadratic upper bound:

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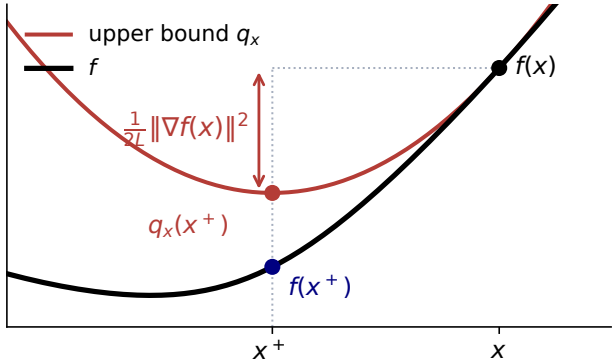
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Guaranteed decrease with $t = 1/L$

$$f(x^+) \leq q_x(x^+) = f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$$

Optimizing the upper bound: picture



x^+ minimizes q_x; since $f \leq q_x$, f drops by at least $\frac{1}{2L}\|\nabla f(x)\|^2$

Q: A safe step decreases f; what makes progress slow?

Strong convexity: positive curvature

Definition (Strongly convex)

A twice continuously differentiable $f : \mathbb{R}^n \to \mathbb{R}$ is μ-**strongly convex**, $\mu > 0$, if for every x,

$$\nabla^2 f(x) \succeq \mu I.$$

all Hessian eigenvalues are at least μ: curvature is uniformly positive

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(proof: Taylor's theorem again)

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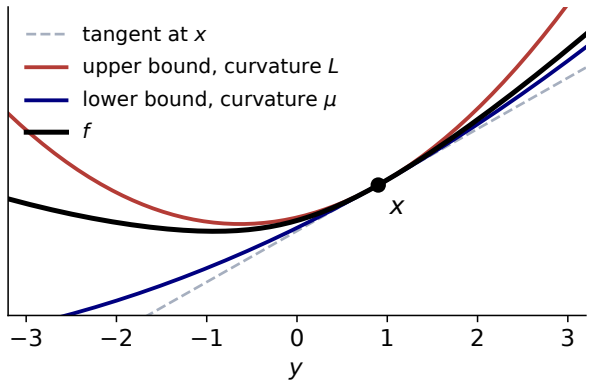
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A: When $A \succ 0$, with $\mu = \lambda_{\min}(A) > 0$.

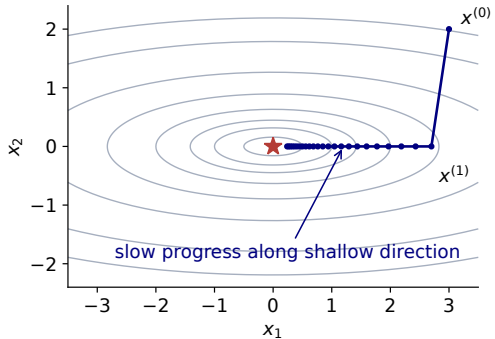
Curvature bounds: picture



both parabolas touch f at x: L-smoothness keeps f below the red one; μ-strong convexity keeps f above the blue one

Quadratics: curvature and progress

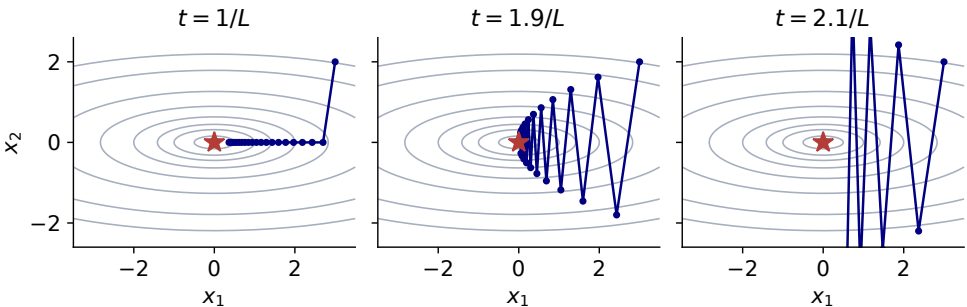
$$f(x) = \tfrac{1}{2} x^T A x, \qquad A = \begin{bmatrix} 1 & 0 \\ 0 & 10 \end{bmatrix} \qquad \Longrightarrow \qquad \mu = 1, \quad L = 10$$



Q: Why does $t = 1/L$ make slow progress along the shallow direction?

Quadratics: choosing the step size

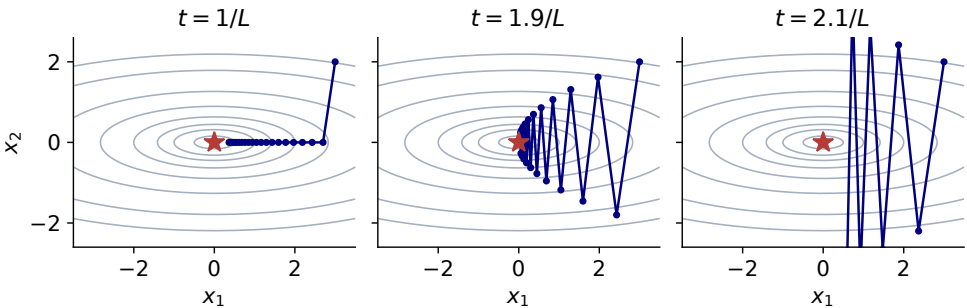
Consider $f(x) = x^T A x$ with A diagonal. Then each coordinate evolves separately: $x_i^{(k+1)} = (1 - t\lambda_i)\, x_i^{(k)}$.



Q: Which fixed t makes the worst factor $\max_i |1 - t\lambda_i|$ smallest?

Quadratics: choosing the step size

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Q: Which fixed t makes the worst factor $\max_i |1 - t\lambda_i|$ smallest?
A: $t = 2/(\mu + L)$, with factor $(L - \mu)/(L + \mu) = 9/11$; for $t > 2/L$, the L-coordinate grows if it starts nonzero.

Some important functions

let $A \in \mathbb{R}^{m \times n}$ have rows a_i^T, and let $b \in \mathbb{R}^m$

- **Quadratic loss.** $f(x) = \|Ax - b\|^2$

$$\nabla^2 f(x) = 2A^T A$$

- **Logistic loss.** $\ell(x) = \sum_{i=1}^m \log(1 + \exp\left(-b_i a_i^T x\right))$, $b_i \in \{-1, 1\}$

$$\nabla^2 \ell(x) = A^T D(x) A, \qquad 0 \prec D(x) \preceq \tfrac{1}{4} I$$

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Q: Which are smooth? Which are strongly convex?
A: Both are smooth: $L = 2\|A\|_2^2$ and $L = \|A\|_2^2/4$ (if $A \neq 0$).
Quadratic loss: strongly convex iff A has full column rank.
Logistic loss: full column rank gives strong convexity on bounded convex sets, but not on all of $\mathbb{R}^n$.

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Analysis via Polyak-Lojasiewicz condition

The Polyak-Lojasiewicz condition

Definition (Polyak-Lojasiewicz condition)

A differentiable $f : \mathbb{R}^n \to \mathbb{R}$ is μ-**PL** if for some $\mu > 0$ and every x,

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu(f(x) - f^\star), \qquad f^\star = \inf_x f(x) > -\infty.$$

when we are far from the optimal value, the gradient cannot be too small

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why use PL?

- weaker than strong convexity: allows flat directions and nonunique minimizers
- does not require convexity
- combines directly with guaranteed decrease to prove a linear rate

PL with flat directions

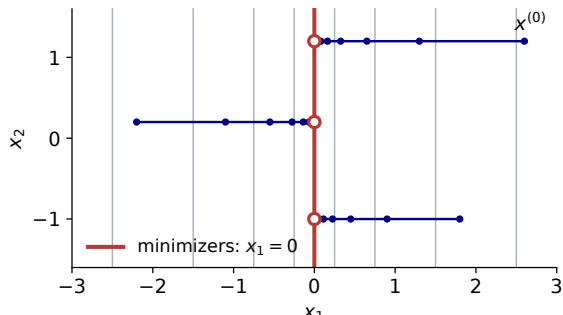
Theorem ([Karimi, Nutini, and Schmidt (2016)])

If $f(x) = g(Ax)$, where $g : \mathbb{R}^m \to \mathbb{R}$ is strongly convex and A is linear, then f satisfies PL.

example: least squares, $f(x) = \|Ax - b\|^2$, even if A is rank deficient

PL with flat directions: picture

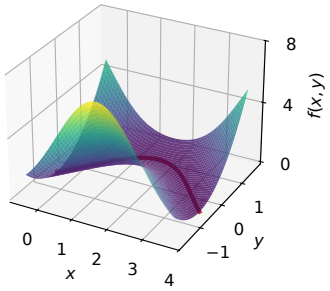
$f(x) = 5x_1^2$, GD with $t = \frac{1}{2L}$: $x_1^{(k+1)} = \frac{1}{2}x_1^{(k)}$, $x_2^{(k+1)} = x_2^{(k)}$

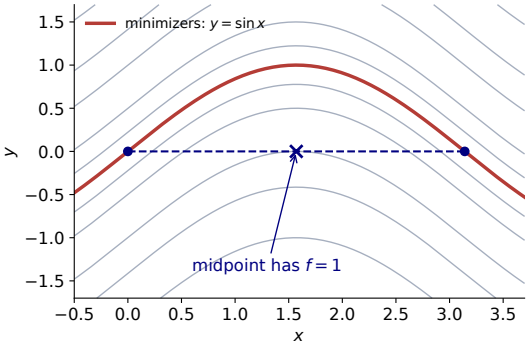


the gradient never moves along the flat direction: each run converges to the minimizer $(0, x_2^{(0)})$, which depends on where it starts

River valley

$$f(x, y) = (y - \sin x)^2, \qquad f^\star = 0$$





Q: Can this nonconvex function satisfy PL?

River valley satisfies PL

write $r = y - \sin x$, so $f(x, y) = r^2$ and

$$\nabla f(x, y) = 2r \begin{bmatrix} -\cos x \\ 1 \end{bmatrix}.$$

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$$\frac{1}{2}\|\nabla f(x, y)\|^2 = 2r^2(1 + \cos^2 x) \geq 2f(x, y)$$

- f is 2-PL, with a whole curve of minimizers $y = \sin x$
- f is not convex: two points on the curve can have a midpoint above it
- f is **not globally smooth** but smooth on any compact set

Optimality condition for PL function

Theorem

Any stationary point of a Polyak-Lojasiewicz function is globally optimal.

Optimality condition for PL function

Theorem

Any stationary point of a Polyak-Lojasiewicz function is globally optimal.

proof: if $\nabla f(\bar{x}) = 0$, then

$$0 = \frac{1}{2}\|\nabla f(\bar{x})\|^2 \geq \mu(f(\bar{x}) - f^\star) \geq 0$$

$\implies$ $f(\bar{x}) = f^\star$ is the global optimum.

strong convexity $\implies$ Polyak-Lojasiewicz

Theorem

If f is μ-strongly convex, then f is μ-Polyak-Lojasiewicz.

strong convexity $\implies$ Polyak-Lojasiewicz

Theorem

If f is μ-strongly convex, then f is μ-Polyak-Lojasiewicz.

proof: minimize the strong convexity condition over y:

$$\begin{aligned} \min_y f(y) &\geq \min_y \left(f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2} \|y - x\|^2 \right) \\ f^\star &\geq f(x) - \frac{1}{2\mu} \|\nabla f(x)\|^2 \end{aligned}$$

since $y = x - \nabla f(x)/\mu$ minimizes the quadratic lower bound

Objective convergence and iterate convergence

- **objective convergence:** $f(x^{(k)}) \to f^\star$
- **iterate convergence:** $x^{(k)} \to \bar{x}$ for some minimizer $\bar{x}$

under **strong convexity**, objective convergence implies iterate convergence:

$$f(x^{(k)}) - f^\star \geq \frac{\mu}{2}\|x^{(k)} - x^\star\|^2$$

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under **PL alone**, this implication fails for an arbitrary sequence:

$$f(x_1, x_2) = x_1^2, \qquad x^{(k)} = (0, (-1)^k).$$

all objective values are optimal, but these points do not converge

this is not a GD trajectory: with a zero gradient, GD stays put

Rates of convergence

write $e_k = f(x^{(k)}) - f^\star$ and suppose $e_0 > 0$

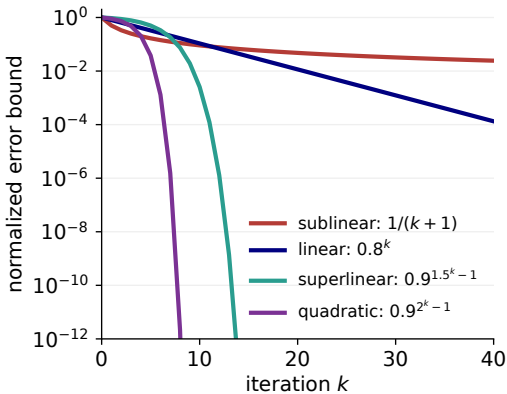
sublinear example: $e_k = O(1/k)$
flattens out on semilog axes

linear: $e_k \le c^k e_0$, $0 < c < 1$
straight on semilog axes

superlinear: $e_{k+1}/e_k \to 0$
bends downward on semilog axes

quadratic: $e_{k+1} \le C e_k^2$
correct digits roughly double each step
(Newton's method, later)

GD with $t = 1/L$ on smooth convex functions has an $O(1/k)$ guarantee.
Smooth PL gives a linear rate.



Gradient descent converges linearly

Theorem

If $f : \mathbb{R}^n \to \mathbb{R}$ is μ-Polyak-Lojasiewicz, L-smooth, and a minimizer exists, then gradient descent with step size $1/L$

$$x^{(k+1)} = x^{(k)} - \frac{1}{L}\nabla f(x^{(k)})$$

has objective error $f(x^{(k)}) - f^\star \leq (1 - \mu/L)^k (f(x^{(0)}) - f^\star)$.

Gradient descent converges linearly: proof

write $e_k = f(x^{(k)}) - f^\star$

smoothness + the update:

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{1}{2L}\|\nabla f(x^{(k)})\|^2$$

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PL:

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combine the two:

$$e_{k+1} \leq e_k - \frac{\mu}{L} e_k = \left(1 - \frac{\mu}{L}\right) e_k \leq \left(1 - \frac{\mu}{L}\right)^{k+1} e_0$$

objective decrease proportional to remaining error $\implies$ linear convergence

How many iterations?

let $\kappa = L/\mu$, and suppose $0 < \mu < L$ and $e_0 = f(x^{(0)}) - f^\star > \varepsilon > 0$

$$e_k \leq \left(1 - \frac{1}{\kappa}\right)^k e_0 \leq \exp\left(-k/\kappa\right) e_0$$

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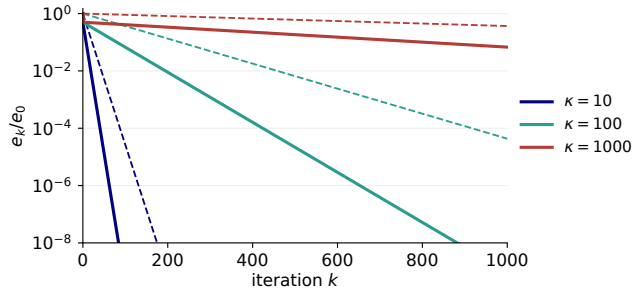
therefore it suffices to take

$$k \geq \kappa \log \frac{e_0}{\varepsilon}.$$

- larger curvature ratio κ means a slower guarantee
- for a positive definite quadratic, $\kappa = \lambda_{\max}(A)/\lambda_{\min}(A)$
- if $\mu = L$, the bound gives $e_1 = 0$: one step suffices

How many iterations? In practice

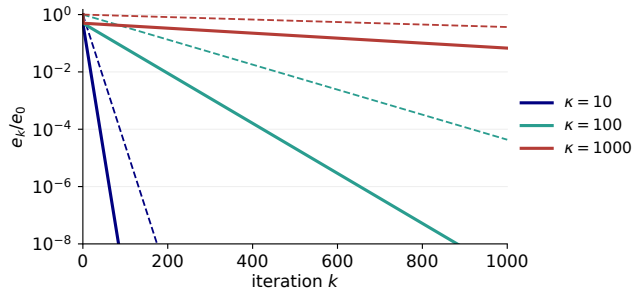
GD with $t = 1/L$ on $\frac{1}{2}x^T \mathbf{diag}(1, \kappa)x$. solid: measured; dashed: bound $(1 - 1/\kappa)^k$



$10\times$ larger $\kappa \implies 10\times$ flatter slope (true rate here: $(1 - 1/\kappa)^2$)

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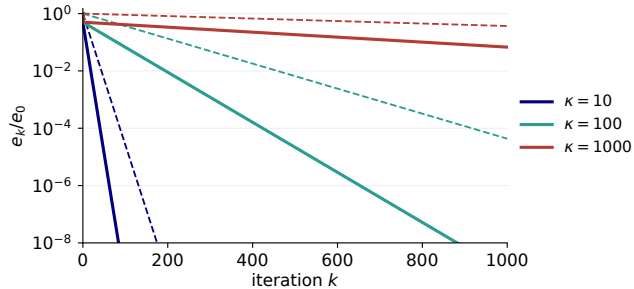


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Q: For $\kappa = 1000$, how many iterations does the bound need to reach $e_k/e_0 \leq 10^{-6}$?

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Q: For $\kappa = 1000$, how many iterations does the bound need to reach $e_k/e_0 \leq 10^{-6}$?

A: $k \geq \kappa \log 10^6 \approx 1000 \times 13.8 \approx 14{,}000$.

Practical convergence

exact line search achieves at least the same decrease from the current x:

$$f(x^+) = \min_{t \geq 0} f(x - t\nabla f(x)) \leq f\left(x - \frac{1}{L}\nabla f(x)\right)$$

when the line minimum exists; the same linear-rate bound follows

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local vs global: the constants must hold wherever the proof uses them

- global smoothness + global PL: guarantee from any starting point
- local bounds: first justify that the iterates stay in the relevant region